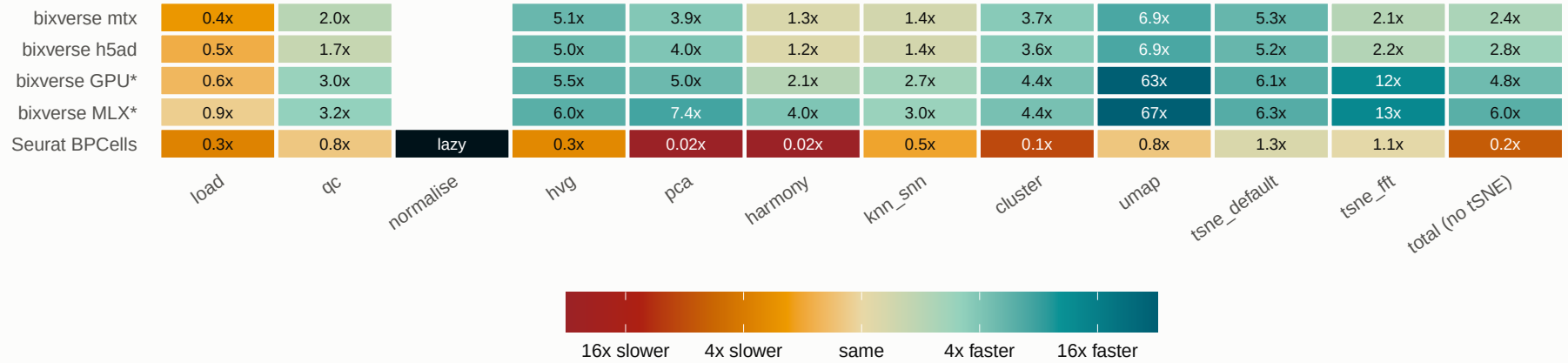
## Speed−up per step against Scanpy with Dask, full data set

In−memory Scanpy dies at load here; Dask is the Scanpy path that finishes. Only runs that completed.




bixverse normalises during load; Dask defers normalisation, so the normalise column carries no fold change.
Dask has no DGE step (Wilcoxon fails on Dask arrays), so its total lacks one; in−memory Scanpy's takes 602 s at 500k.
bixverse CPU tSNE builds its kNN with IVF.